

ITA0506-COMPUTER VISION LAB EXPERIMENT

DATE :18-07-2026

EXPNO:8

AIM: Scaling an image to its Bigger and Smaller sizes.

Algorithm:

- Import the required libraries, mainly cv2, and read the input image using cv2.imread().
- Check the image size using image.shape so you know the original width and height before scaling.
- For **bigger size**, resize the image using scale factors greater than 1, or give a larger width and height using cv2.resize().
- For **smaller size**, resize the image using scale factors less than 1, or give a smaller width and height using cv2.resize().
- Display or save the resized images, using suitable interpolation such as cv2.INTER_CUBIC for enlarging and cv2.

Code:

```
import cv2
mode = input("Enter mode (normal/slow/fast): ").lower()
if mode == "slow":
    delay = 100
elif mode == "fast":
    delay = 5
else:
    delay = 25
cap = cv2.VideoCapture(0)
if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()
while True:
    ret, frame = cap.read()
```

```

if not ret:
    print("Error: Could not read frame.")
    break
cv2.imshow("Webcam Video", frame)
if cv2.waitKey(delay) & 0xFF == ord('q'):
    break

```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

OUTPUT :

